Tactical adaptation: population outcomes

Ten independent pilots • 150 additional epochs each • all training runs completed

38.6%

CLEAN WINS

32.7% before / +5.9 points

83.8%

SURVIVAL

75.1% before / +8.7 points

10 / 10

PILOTS COMPLETED

150 epochs each / 0 worker retries

Summary of findings

On the same separate tactical cases, the adapted pilots won more often and survived more often.
Mutual destruction increased slightly: 9.1% → 10.0%.

Experimental development

Independent population training, controlled PPO experiments and tactical adaptation

04 → 05	One PPO pilot became an arbitrary-size population with separate rewards and checkpoints.
Reward diversity	Monotonic sweep → relative Halton profiles → seeded, independent uniform ranges.
Evaluation design	Recorded training/validation and a fixed benchmark; final testing after development.
Optimisation	Lower learning rate and selected-skill probabilities improved clean wins.
Bounded trials	More parameter heads underperformed; the terminal-bonus trial stopped incomplete.
Tactical	The stronger shared-head population was retained and adapted to scripted tactical.

Scenario collection and PPO updates

75 scenarios per epoch for each pilot; shared scenario schedules support matched comparisons

OBSERVE

20 frames × 51 features
2 seconds of flight history

→

CHOOSE

One of 33 skills
+ its numeric parameters

→

SIMULATE

Execute about 1 second
Record events and rewards

→

UPDATE

Estimate advantages
PPO minibatches of 256

Scenarios and minibatches

A scenario is one duel with seeded starting geometry, speeds and altitudes (up to 120 seconds).
75 scenarios produce many decision transitions. A PPO minibatch contains 256 transitions.
150 epochs × 75 scenarios = 11,250 training flights per pilot; 112,500 across the population.

Actor–critic network architecture

Shared transformer: width 64 • 4 attention heads • 2 layers • shared parameter predictor

ACTOR: policy distribution

Predict a distribution over 33 skills
and 16 numeric parameter slots.
Only parameters used by the chosen
skill count in its PPO probability.

CRITIC: value estimation

Predict future discounted reward.
Compare rewards + future value with
that prediction to estimate advantage.
Positive advantage favours the action.

$$\log \pi(a \mid s) = \log \pi(k \mid s) + \sum_{j \in \text{active}(k)} \log p(z_j \mid s)$$

The selected-skill correction changed the learning objective, without adding network capacity.

Reward profiles and common evaluation metrics

Each pilot draws its coefficients once using a reproducible random seed (42)

Crash / being shot down	Each independently uniform: −450 to −300
Lock / launch with lock	Each independently uniform: 0 to 2
Avoid an incoming missile	Uniform: 0 to 15
Launch without lock	Uniform: −12 to −4
Opponent destroyed / clearance	Fixed: +250 / +0.02

Common outcome scoring

Clean win +100 • survive without elimination 0 • mutual destruction −50 • own loss −100

Shared development benchmark

Simple opponent • the same 10 reward profiles and 100 cases per pilot • seed 211307

Population	Pilots / cases	Clean wins	Status
Original	10 / 100	28.2%	Starting population
Reduced learning rate	10 / 100	36.8%	Retained
Selected-skill probabilities	10 / 100	50.0%	Tactical parent
Per-skill parameter heads	10 / 100	42.5%	Not retained

Bonus trial: 6 completed pilots only; matched development clean wins 51.5% → 49.8%.

Fixed development cases support comparison across runs; they are not an untouched final test.

Recorded training and validation

Population mean own return • identical reward profiles across runs • 150 epochs per pilot



Validation selects checkpoints. Own-return curves describe learning; clean wins measure combat outcomes.

Tactical adaptation: experimental design

Same reward profiles, skills, physics and PPO settings; new scripted opponent behaviour

BEFORE

Freeze the 10 selected-skill parents.
Assess all on 300 tactical cases.
Keep the originals unchanged.

FINE-TUNE

Load each pilot's actor + critic.
Fresh Adam; LR 0.00003.
150 epochs × 75 fresh tactical flights.

SELECT

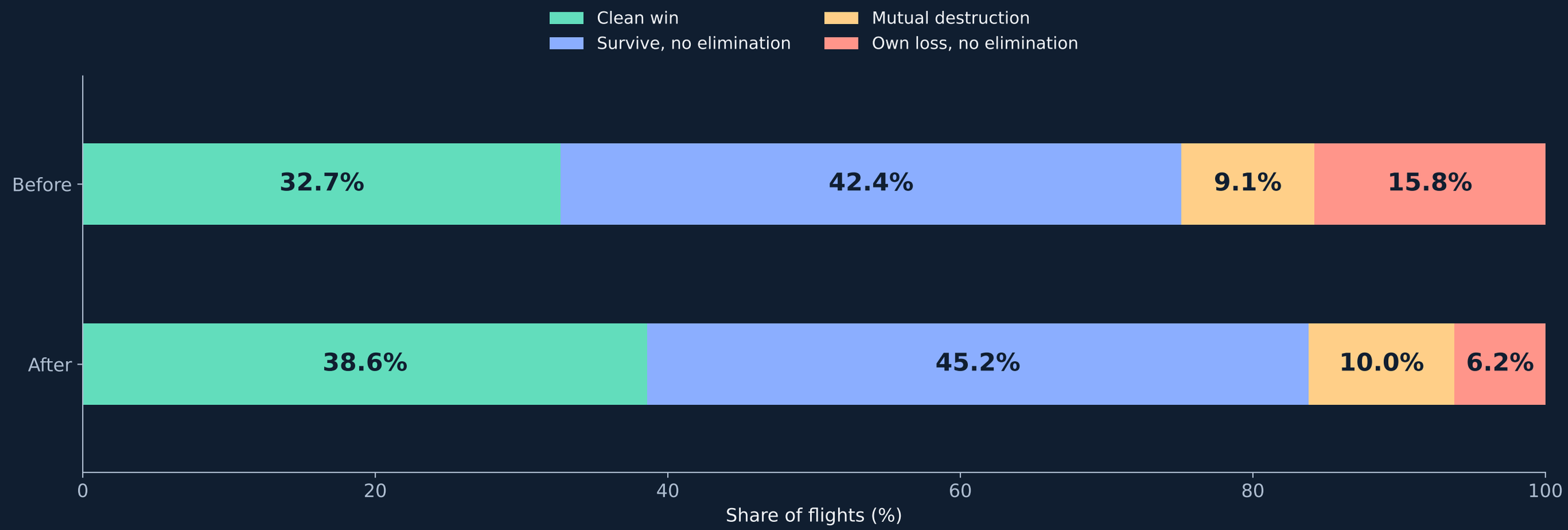
50 separate validation cases.
Own reward selects the checkpoint;
initial weights are also eligible.

COMPARE

Repeat the same 300-case assessment.
Use a separate 100-case confirmation
for both parent and adapted pilots.

Tactical confirmation outcomes

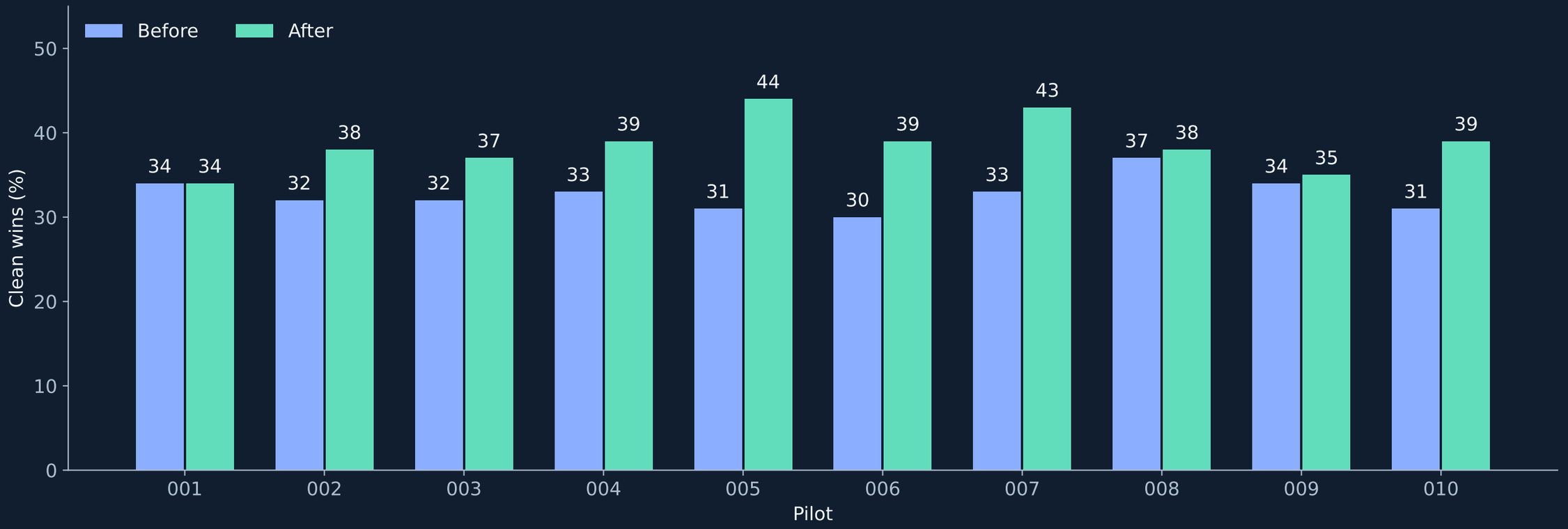
Separate 100-case tactical confirmation • population means across all ten pilots



Clean-win gain: +5.9 pp | Paired scenario bootstrap 95% interval: +1.2 to +11.0 pp

Per-pilot confirmation results

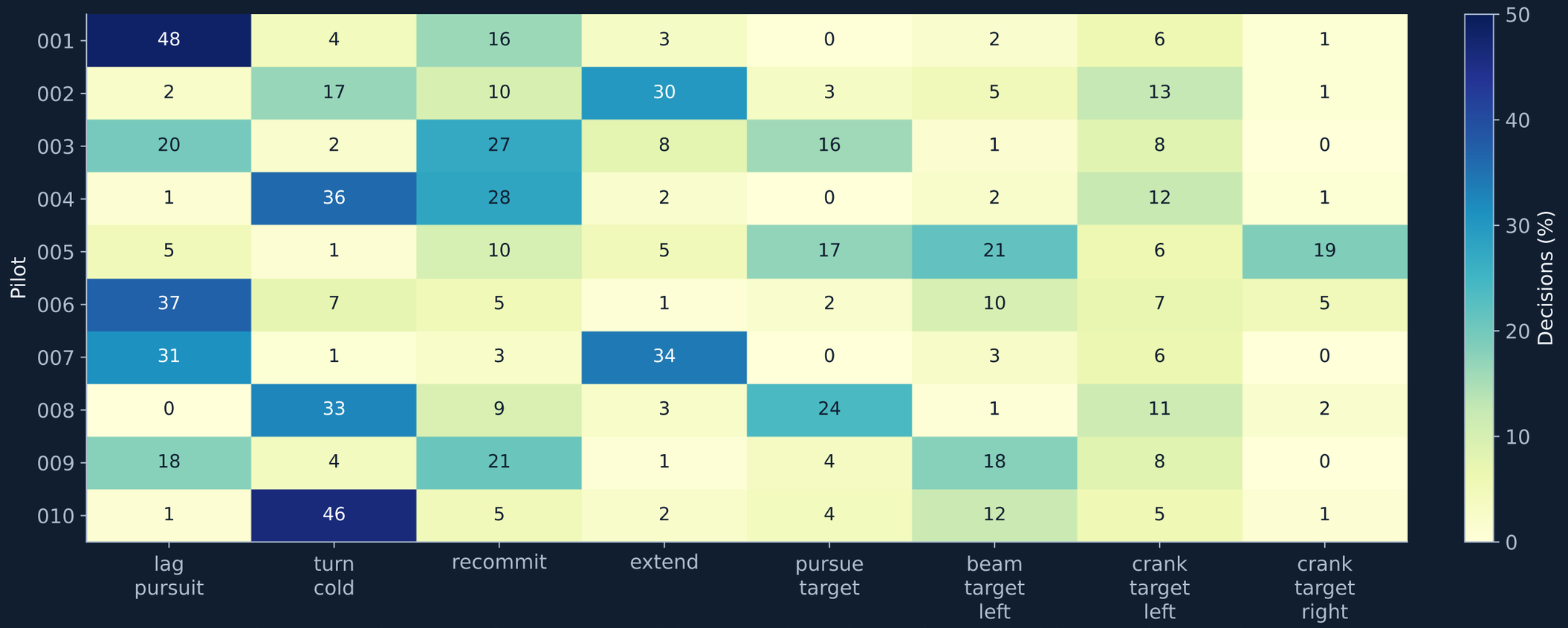
Nine clean-win rates improved and one tied on the same 100 tactical confirmation cases



Pilot 005: 44% clean wins, 86% survival. Pilot 007: 43%, 85%. Small gaps are uncertain.

Behavioural variation after adaptation

Observed decision frequencies on confirmation flights • descriptive measures, not aggression rankings



001 favours lag pursuit; 010 often turns cold; 005 mixes beaming, cranking and pursuit.

Generalisation and evaluation limitations

Matched performance improvements and the scope of the available evidence

OBSERVED RESULTS

100-case confirmation: +5.9 pp clean wins.
300-case assessment: +6.9 pp clean wins.
Survival improved in both suites.
All 10 training runs completed; parents and frozen source files verified unchanged.

EVALUATION LIMITATIONS

One training seed per reward profile.
No post-adaptation test against simple.
Mutual destruction did not improve on the confirmation suite.
No universal best pilot or style proven.

No direct seed overlap was found across training, selection and tactical report suites.

Held-out gains do not establish the absence of overfitting to this simulator or opponent.

Portable reports and supporting data

PDF and self-contained HTML formats; no notebook runtime, Python, GPU or LFS download required

REPORTS

presentation.pdf — slide deck
presentation.html — offline browser deck
README.md — technical report

DATA

data/results.json — verified summaries
data/learning_curves.csv — recorded epochs
data/development_summary.csv — pilot results

Principal findings

Controlled comparisons improved PPO performance in a diverse pilot population. Subsequent tactical adaptation improved clean wins and survival on separate evaluation cases, with continued variation in observed behaviour.